<u>DevOps and Automations</u> <u>TASK – 2</u>

1.Install Terraform on your system and verify the installation by running terraform --version in your terminal.

Ans :

- Download Terraform
- Go to the official Terraform download page:
- Terraform Downloads
- Download the Windows AMD64 ZIP if you're using a normal 64-bit Windows computer.
- Extract Terraform
- Extract the downloaded ZIP file.
- You should get:
- terraform.exe
- For example, put it in:
- C:\Terraform
- So the file should be:
- C:\Terraform\terraform.exe
- Add Terraform to PATH
- Press Windows + S.
- Search for:
- Environment Variables
- Click Edit the system environment variables.
- Click Environment Variables.
- Under System variables, find:
- Path

- Click Edit.
- Click New.
- Add:
- C:\Terraform
- Click OK on all windows.
- Open a new terminal
- Close your existing PowerShell/Command Prompt and open a new one so the PATH change takes effect.
- Verify Terraform
- Run:
- terraform --version

**2.Write a basic Terraform script (main.tf) that provisions an AWS EC2 instance with the t2.micro type in the us-east-1 region.

Hint: Use the aws_instance resource and specify the AMI ID for Amazon Linux 2.**

Ans :

Steps

- Create a folder, for example:
- terraform-ec2
- Inside the folder, create:
- main.tf
- Paste the Terraform code above into main.tf.
- Open Terminal/PowerShell in that folder.
- Initialize Terraform:

- terraform init
- Validate the configuration:
- terraform validate
- Preview what Terraform will create:
- terraform plan
- To actually create the EC2 instance:
- terraform apply
- When prompted:
- Do you want to perform these actions?
- type:
- yes

3.Add a tag to your EC2 instance in your Terraform script so that it displays your name as the 'Owner' in the AWS console.

Ans :

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
    }
  }
}


provider "aws" {
  region = "us-east-1"
```

```
}

resource "aws_instance" "example" {
  ami           = "ami-0c02fb55956c7d316"
  instance_type = "t2.micro"

  tags = {
    Name  = "Terraform-EC2"
    Owner = "YOUR_NAME"
  }
}
```

4.Initialize and apply your Terraform configuration to create the EC2 instance, then verify in the AWS console that your instance is running.

Ans :

- Open PowerShell/Terminal.
- Go to the folder containing your main.tf:
- cd C:\terraform-ec2
- Initialize Terraform:
- terraform init
- Check the configuration:
- terraform validate
- Preview the resources Terraform will create:
- terraform plan

- Create the EC2 instance:
- terraform apply
- When Terraform asks:
- Do you want to perform these actions?
- type:
- yes
- Wait until Terraform displays:
- Apply complete!
- Open the AWS EC2 Console.
- Make sure the AWS Region is:
- US East (N. Virginia) / us-east-1
- In the left navigation menu, click Instances → Instances.
- Find your instance named:
- Terraform-EC2
- Select the instance.
- Check the Instance state. It should show:
- Running
- Check the Instance type. It should show:
- t2.micro
- Open the Tags tab and verify:
- Name   Terraform-EC2
- Owner YOUR_NAME

- Check Status check. After the instance has had time to initialize, the checks should pass. AWS documents the EC2 console's Status check column and the Status and alarms tab for reviewing these checks. AWS Documentation+1
- Take a screenshot showing:
- Terraform-EC2
- Instance state: Running
- Instance type: t2.micro

**5.Use ChatGPT or GitHub Copilot to help you write a Terraform script that creates a security group allowing inbound SSH (port 22) access, and add it to your EC2 instance.

Hint: Ask the AI tool for a minimal example with comments, then adapt it to your main.tf file.**

Ans :

- Replace your main.tf with this
- terraform {
- required_providers {
- aws = {
- source = "hashicorp/aws"
- }
- }
- }

- provider "aws" {
- region = "us-east-1"
- }

- # Security group that allows SSH access
- resource "aws_security_group" "ssh_access" {
- name   = "allow-ssh"
- description = "Allow SSH inbound traffic"

- # Allow SSH from the internet
- ingress {
- description = "SSH"
- from_port   = 22
- to_port   = 22
- protocol   = "tcp"
- cidr_blocks = ["0.0.0.0/0"]
- }

- # Allow all outbound traffic
- egress {
- from_port   = 0
- to_port   = 0
- protocol   = "-1"
- cidr_blocks = ["0.0.0.0/0"]
- }

- tags = {
- Name = "SSH-Security-Group"
- Owner = "YOUR_NAME"
- }
- }

- # EC2 instance
- resource "aws_instance" "example" {
- ami   = "ami-0c02fb55956c7d316"
- instance_type = "t2.micro"

- # Attach the SSH security group to the EC2 instance
- vpc_security_group_ids = [aws_security_group.ssh_access.id]

- tags = {
- Name = "Terraform-EC2"
- Owner = "YOUR_NAME"
- }
- }
- Replace YOUR_NAME with your actual name.
- Initialize Terraform
- Open PowerShell in your Terraform folder:
- terraform init
- Format the file
- terraform fmt
- Validate the configuration
- terraform validate
- You should get:
- Success! The configuration is valid.
- Preview the changes
- terraform plan
- Check that Terraform plans to create/update the security group and attach it to the EC2 instance.
- Apply the configuration
- terraform apply

- When prompted, enter:
- yes